Draft Analysis Report

1. Scenario Description

The Control Group Consortium is a collaborative organization in which universities, hospitals, and research institutions share research facilities, equipment, and technical staff. The proposed database supports the coordination of these resources across institutions and locations, connecting research activities with scheduling, access requirements, actual usage, and financial responsibility.

The system is analyzed by following how a research need becomes a scheduled activity and then a billable cost. Researchers participate in projects, and each project may include multiple experiments. For an experiment, a researcher submits a request specifying resource types, preferred times, and support needs. Consortium staff review the request and record their decision, which may involve an alternative arrangement. Scheduled activities are represented as events linked to specific resources, facility configurations, and staff assignments. Events also represent maintenance and calibration, allowing these activities to be considered alongside research sessions when assessing availability.

Scheduling alone does not describe actual resource use or establish permission to use a resource. The system therefore records usage separately and connects people to institutional affiliations, roles, qualifications, and access permissions. Financial records connect actual usage to applicable rates, funding sources, cost-sharing policies, and invoices. This separation is necessary because requested arrangements may differ from approved schedules, actual usage may differ from planned usage, and the person conducting research may differ from the party responsible for payment.

Based on these relationships, the schema is organized into seven connected areas: Institution and People, Resources, Access Control, Project and Request, Event, Pricing and Allocation, and Billing. Together, these areas provide a traceable record of who requests, accesses, and uses shared resources, how activities are coordinated, and how the resulting costs are assigned.

2. Organizational Schema

2.1 Institution and People

This area records participating institutions, individuals, and payment responsibility. Organizations, their departments, and individual participants are represented by `institution`, `department`, and `person`, respectively. Each department belongs to an institution, while personal information is stored independently of institutional affiliation.

The relationship between a person and an institution is recorded through `affiliation`. Its `validFrom` and `validTo` attributes record when that relationship applies, allowing changes in affiliation to be retained and a person to be associated with multiple institutions without duplicating personal information.

Payment responsibility is represented separately through `payer`, which can refer to a person, institution, or department. This distinction is needed because the person using a resource may differ from the party responsible for its cost.

2.2 Resources

This area describes shared facilities and equipment, their locations, configurations, and qualification requirements. Facilities and equipment are represented by `resource` because both have capabilities, limitations, and usage constraints. Each resource is classified through `resource_type` and associated with a `site,` which identifies its institutional location and time zone.

Equipment location is recorded separately through `equipment_placement`, with start and end times that preserve movement between facilities. Facility arrangements are described by `configuration`. These entities distinguish the resource itself from where it is placed and how a facility is configured.

Qualification requirements are defined through `qualification` and linked to resources by `resource_requirement`. Qualifications held by individuals are recorded through `person_qualification`, including their issuer and validity dates. Separating resource requirements from individual qualifications provides the information needed to assess whether a person is qualified to use a resource.

2.3 Access Control

This area records identity credentials, access permissions, and access history. Organizational roles are defined by `role` and assigned through `person_role`, allowing one person to hold multiple roles. The `canViewConfidential` attribute provides a role-level indicator for access to confidential information.

Identity and permission are recorded separately because an identifiable person is not necessarily authorized to use every resource. The `credential` entity connects a person to an issuing institution and records the credential's validity period. The `access_grant` entity records permission to access a particular resource, together with the authorizing person, validity period, and any associated project.

Actual access attempts are recorded through `access_log`, including the credential, resource, time, result, and reason. This separates permission to access a resource from evidence of whether access occurred or was denied.

2.4 Project and Request

This area organizes research work and records how resource requests are evaluated. A `project` identifies the overall research undertaking and its principal investigator. Each `experiment` belongs to a project and records a specific research activity, its lead researcher, and its time span. Project participants and their responsibilities are connected through `project_member`.

Research needs are recorded separately from the experiment through `request`. Each request identifies its requester and requesting institution, together with preferred dates, staffing needs, and safety or special conditions. The `request_need` entity records the resource types, quantities, and estimated usage required. Requests therefore describe the kinds of resources needed, while specific resources are assigned during event scheduling.

The outcome of request review is recorded through `decision`, including who made the decision, when it was made, and its rationale. Separating requests from decisions preserves the distinction between the researcher's original needs and the arrangements accepted or proposed by consortium staff. A decision can be linked to the scheduled event to which it relates.

2.5 Event

This area records scheduled activities and the resources and personnel assigned to them. The `event` entity provides a common structure for experiment sessions, preparation, maintenance, calibration, and other activities that occupy resources. Its type, start and end times, and status describe the activity's place in the schedule. Multiple events can represent separate sessions of an experiment.

Resources, facility configurations, and supporting personnel are linked to events through `event_resource`, `event_configuration`, and `staff_assignment`. Separate association entities allow an event to involve multiple resources and staff members without repeating the event record. The duty attribute describes each assigned person's responsibility.

Staff working availability is recorded through `staff_availability`, separately from assigned work. Changes to event times and status are retained through `event_change`, including the previous and revised values, the person making the change, and the reason. These records provide information for reviewing availability, identifying scheduling conflicts, and tracing schedule changes.

2.6 Pricing and Allocation

This area defines how usage is priced, which funding sources can cover it, and

how costs are shared. The `rate` entity records prices for resource types, consumables, or staff services. Its `billingUnit` and validity dates specify how and when a price applies. Consumables are described separately through `consumable`, while `estimate` links an event to a rate and an expected quantity to support projected costs before actual usage is known.

Funding sources are represented by `funding_resource`, which records the responsible payer, funding type, budget, and validity period. The `funding_eligibility` entity connects funding sources to the rates they are eligible to cover. This separates the existence of available funding from permission to use it for a particular expense.

Cost-sharing rules are recorded through `allocation_policy,` including the associated project, authorization, rationale, and validity period. Each `policy_share` identifies a funding source, a beneficiary project, and a share expressed through `basisPoint`. Separating a policy from its individual shares allows one policy to distribute costs across multiple funding sources and projects.

2.7 Billing

This area records actual usage, the resulting charges, and their inclusion in invoices. The `usage_record` entity links an event to the resource, staff member, or consumable used, together with the user, quantity, and usage times. Actual usage is stored separately from scheduled events and estimates because it may differ from the original plan.

A `charge` connects a usage record to the applicable rate and allocation policy. Authorized credits or additional fees are recorded through `adjustment`, preserving their amount, reason, and authorization separately from the charge. The `charge_allocation` entity records the amounts assigned to individual policy shares.

Invoices are represented by `invoice`, which identifies the funding source, billing period, issue date, and status. The `invoice_lines` entity connects charge allocations to invoices. Together, these records provide a traceable path from an invoiced amount back to its allocation, charge, actual usage, and scheduled activity.

3. List of Assumptions

- Each department and site belongs to one institution. A person may have multiple institutional affiliations, with the validity period recorded separately for each affiliation.

- Each payer represents only one person, institution, or department.

- Each resource has one resource type and one associated site. Equipment placement records identify the facility where an item is placed during a specified period.

- Each project has one principal investigator and may include multiple members and experiments. Each experiment belongs to one project and has one lead researcher.

- Each request concerns one experiment and identifies one requester and one requesting institution. A request may specify multiple resource types and have multiple recorded decisions.

- Each event may be associated with one experiment or exist independently. An event may involve multiple resources, configurations, and staff members.

- Each access grant applies to one person and one resource, with an optional project association. Each access log identifies one credential and one resource, with an optional event association.

- Each rate applies to only one pricing subject: a resource type, a consumable, or a staff service. Each usage record describes only one usage category.

- Each funding source has one payer and may cover multiple eligible rates. Each allocation policy belongs to one project, while its shares may identify different funding sources and beneficiary projects.

- Each charge refers to one usage record, one rate, and one allocation policy. Multiple adjustments and charge allocations may be associated with the same charge.

- Each invoice is associated with one funding source and may contain multiple charge allocations. Each charge allocation may appear on at most one invoice.